Appendix B: Storage and prediction excerpt

storage/flight_data_store.py

```
import math
import sqlite3
import threading
import csv
import sys
import importlib
import io
from collections import defaultdict
from datetime import datetime, timedelta, timezone
from pathlib import Path

def _try_import_openpyxl():
    try:
        mod = importlib.import_module("openpyxl")
        return getattr(mod, "Workbook"), getattr(mod, "load_workbook"), True
    except ModuleNotFoundError:
        pass

    # Fallback: project-local virtualenv site-packages
    project_root = Path(__file__).resolve().parents[1]
    venv_site = project_root / "venv" / "Lib" / "site-packages"
    if venv_site.exists():
        venv_site_str = str(venv_site)
        if venv_site_str not in sys.path:
            sys.path.insert(0, venv_site_str)
        try:
            mod = importlib.import_module("openpyxl")
            return getattr(mod, "Workbook"), getattr(mod, "load_workbook"), True
        except ModuleNotFoundError:
            pass

    return None, None, False


Workbook, load_workbook, _OPENPYXL_AVAILABLE = _try_import_openpyxl()


def _now_iso():
    return datetime.now(timezone.utc).isoformat()


def _parse_ts(value):
    if value in (None, ""):
        return _now_iso(), None
    text = str(value).strip()
    if not text:
        return _now_iso(), None
    try:
        dt = datetime.fromisoformat(text.replace("Z", "+00:00"))
        if dt.tzinfo is None:
            dt = dt.replace(tzinfo=timezone.utc)
        return dt.isoformat(), dt
    except ValueError:
        try:
            dt = datetime.strptime(text, "%Y-%m-%d %H:%M:%S").replace(tzinfo=timezone.utc)
            return dt.isoformat(), dt
        except ValueError:
            return _now_iso(), None


def _to_int(value, default=0):
    try:
        return int(round(float(value)))
    except (TypeError, ValueError):
        return default


def _to_float(value, default=0.0):
    try:
        return float(value)
    except (TypeError, ValueError):
        return default


```